THE NATIONAL UNIVERSITY OF LESOTHO

Department of Mathematics and Computer Science

CS3520 - Computer Organisation and Architecture I

Tutorial 3

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1. A set-associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses.
2. A two-way set-associative cache has lines of 16 bytes and a total size of 8 kB. The 64-MB main memory is byte addressable. Show the format of main memory addresses.
3. Consider a 32-bit microprocessor that has an on-chip 16-kB four-way set-associative cache. Assume that the cache has a line size of four 32-bit words. Draw a block dia- gram of this cache showing its organization and how the different address fields are used to determine a cache hit/miss. Where in the cache is the word from memory location ABCDE8F8 mapped?
4. Consider a dynamic RAM that must be given a refresh cycle 64 times per ms. Each refresh operation requires 150 ns; a memory cycle requires 250 ns. What percentage of the memory's total operating time must be given to refreshes?
5. In this exercise we look at memory locality properties of matrix computation. The following code is written in C, where elements within the same row are stored contiguously. Assume each word is a 64-bit integer.

```
for (I=0; I<8; I++)
    for (J=0; J<8000; J++)
        A[I][J]=B[I][0]+A[J][I];
```

   (a) How many 64-bit integers can be stored in a 16-byte cache block?
   (b) Which variable references exhibit temporal locality?
   (c) Which variable references exhibit spatial locality?

6. Locality is affected by both the reference order and data layout. The same computation can also be written below in Matlab, which differs from C in that it stores matrix elements within the same column contiguously in memory.

```
for I=1:8
    for J=1:8000
        A(I,J)=B(I,0)+A(J,I);
    end
end
```

   (a) Which variable references exhibit temporal locality?
   (b) Which variable references exhibit spatial locality?
   (c) How many 16-byte cache blocks are needed to store all 64-bit matrix elements being referenced using Matlab's matrix storage? How many using C's matrix storage? (Assume each row contains more than one element.)

7. Consider a magnetic disk drive with 8 surfaces, 512 tracks per surface, and 64 sectors per track. Sector size is 1 kB. The average seek time is 8 ms, the track-to-track access time is 1.5 ms, and the drive rotates at 3600 rpm. Successive tracks in a cylinder can be read without head movement.

   (a) What is the disk capacity?

   (b) What is the average access time? Assume this file is stored in successive sectors and tracks of successive cylinders, starting at sector 0, track 0, of cylinder i.

   (c) Estimate the time required to transfer a 5-MB file.

   (d) What is the burst transfer rate?

8. Caches are important to providing a high-performance memory hierarchy to processors. Below is a list of 64-bit memory address references, given as word addresses.

0x03, 0xb4, 0x2b, 0x02, 0xbf, 0x58, 0xbe, 0x0e, 0xb5, 0x2c, 0xba, 0xfd

   (a) For each of these references, identify the binary word address, the tag, and the index given a direct-mapped cache with 16 one-word blocks. Also list whether each reference is a hit or a miss, assuming the cache is initially empty.

   (b) For each of these references, identify the binary word address, the tag, the index, and the offset given a direct-mapped cache with two-word blocks and a total size of eight blocks. Also list if each reference is a hit or a miss, assuming the cache is initially empty.

9. By convention, a cache is named according to the amount of data it contains (i.e., a 4 KiB cache can hold 4 KiB of data); however, caches also require SRAM to store metadata such as tags and valid bits. For this exercise, you will examine how a cache's configuration affects the total amount of SRAM needed to implement it as well as the performance of the cache. For all parts, assume that the caches are direct-mapped, byte addressable, and that addresses and the words are 32 bits.

   (a) Calculate the total number of bits required to implement a 32 KiB cache with two-word blocks.

   (b) Calculate the total number of bits required to implement a 64 KiB cache with 16-word blocks. How much bigger is this cache than the 32 KiB cache described in (a)? (Notice that, by changing the block size, we doubled the amount of data without doubling the total size of the cache.)

   (c) Explain why this 64 KiB cache, despite its larger data size, might provide slower performance than the first cache.

   (d) Generate a series of read requests that have a lower miss rate on a 32 KiB two-way set associative cache than on the cache described in (a).

10. For a direct-mapped cache design with a 32-bit address, the following bits of the address are used to access the cache.

tag	**index**	**offset**
63-10	9-5	4-0

   (a) What is the cache block size (in words)?

   (b) How many blocks does the cache have?

   (c) What is the ratio between total bits required for such a cache implementation over the data storage bits?

   (d) A set-associative cache has a block size of four 16-bit words and a set size of 2. The cache can accommodate a total of 4096 words. The main memory size that is cacheable is 64K * 32 bits. Design the cache structure and show how the processor's addresses are interpreted.